



IILM UNIVERSITY

30 DAYS SUMMER INTERNSHIP OF C++

Application Development & File Management Systems

Student Name: DAKSH SHARMA

Enrollment Number: 25SCS1003004699

Program & Semester: B.Tech CSE, Semester 2

Organization: VIRTUAL WORKS LAB

Official Credentials & Offer



VERIFICATION
DATE: 4 October 2026
REF: VW-SIP-INT0000726-965
MEMO: 007040400-26-100140

CERTIFICATE OF COMPLETION

This is to proudly certify that

DAKSH SHARMA

has successfully completed an intensive internship from 04/06/2026 to 04/10/2026 in the domain of

C++ Programming

During this period, they demonstrated excellent technical skills, dedication, and a strong commitment to delivering high-quality real-world project assignments.



Utkarsh Mishra
CEO

Prince Gaur
CHRO



PERFORMANCE EVALUATION REPORT

Official Record & Assessment

I. INTERNSHIP DETAILS

Intern Name:	DAKSH SHARMA
Internship ID:	VW-SIP-INT0000726-965
Domain:	C++ Programming
Duration:	From 04/06/2026 to 04/10/2026
Total Hours:	160 Hours

II. EVALUATION MATRIX

Assessment Criteria	Score (Out of 100)
Technical Competence & Skills	97
Task Completion & Timeliness	95
Problem Solving & Logic	93
Communication & Documentation	94
Consistency & Effort	96
FINAL SCORE:	95/100
OVERALL GRADE:	A (Excellent)

III. PROJECT / TASK RECORD

Project Title / Task Description	Grade	Authoritative Proof URL
Contact Management System	A	https://github.com/DakshSharma04062026/crm-system
Task Management System	A	https://github.com/DakshSharma04062026/task-mgmt-M
Tic Tac Toe Game	A	https://github.com/DakshSharma04062026/tic-tac-toe
Number Guessing Game	A	https://github.com/DakshSharma04062026/number-guess
Basic Calculator	A	https://github.com/DakshSharma04062026/basic-calculator

IV. REVIEWER REMARKS

"Demonstrated brilliant technical aptitude and a highly proactive approach in learning. The completed projects are of exemplary standard quality, reflecting a deep understanding of the domain."

EVALUATED BY:


Prince Gaur
Program Director & CHRO



NSRF REGISTRATION
UDYAM-SH-26-1169185

To,
DAKSH SHARMA
ILM UNIVERSITY

Date: 4 September 2026
Ref: VW-SIP-INT0000726-965

OFFER LETTER: C++ PROGRAMMING

Dear DAKSH,

Congratulations! We are thrilled to officially offer you an internship position as a C++ Programming at VirtualWorks by Nirma.

During our selection process, your skills, enthusiasm, and technical aptitude stood out. We believe your background will be a fantastic addition to our team, and we are excited to see the innovative ideas you will bring to our upcoming projects.

ROLE DETAILS

Position: C++ Programming

Duration: 04/09/2026 to 04/10/2026

Mode: Remote

EXPECTATIONS

- Complete assigned tasks
- Maintain high quality
- Adhere to deadlines

At VirtualWorks, we are committed to fostering innovation and providing our interns with hands-on experience in building real-world solutions. You will collaborate with our core team, expand your technical skill set, and contribute to impactful projects.

We are incredibly excited to welcome you to Team Nirma!



Utkarsh Mishra
CEO

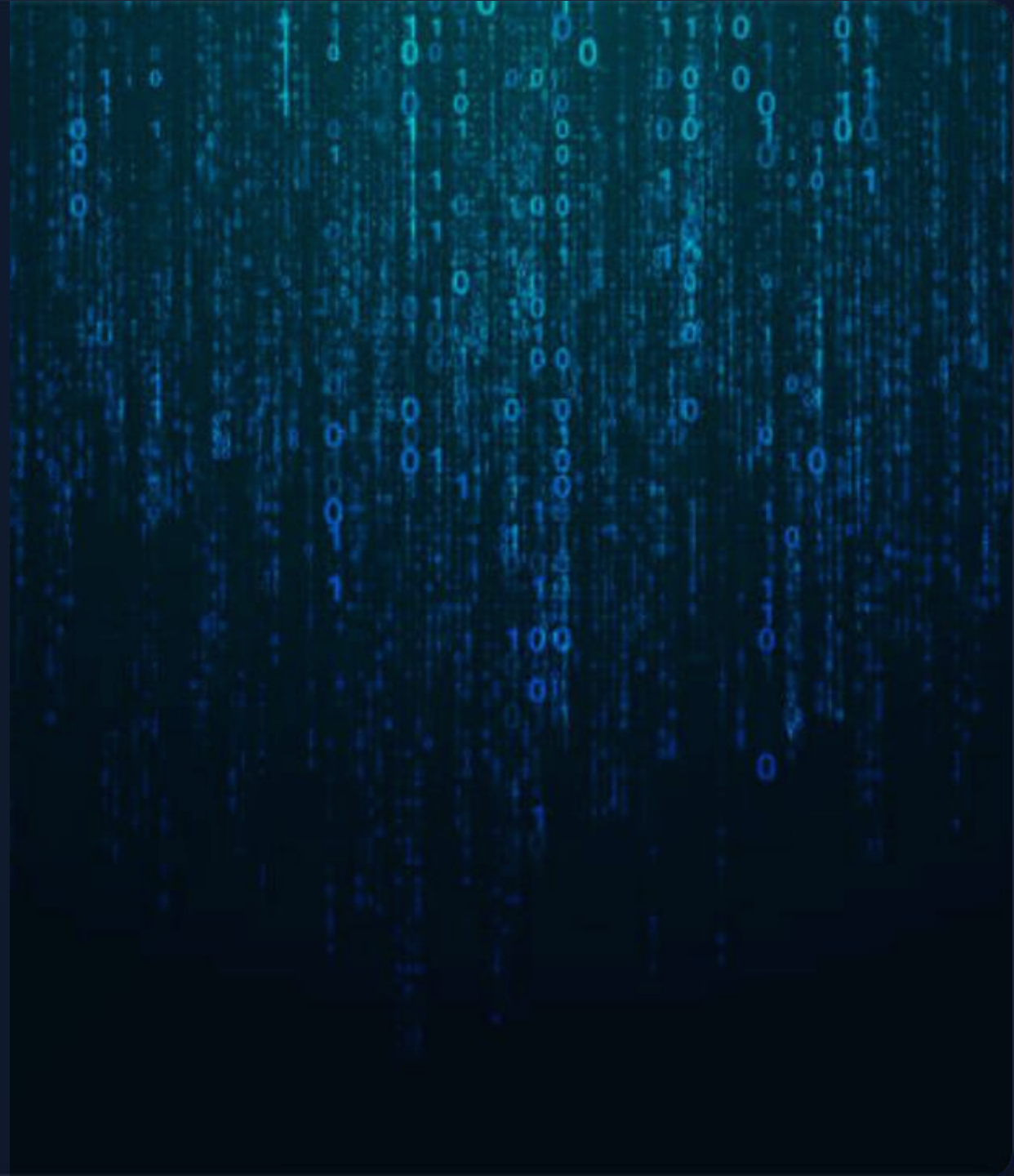
Prince Gaur
CHRO

Internship Overview

From Logic to Persistence

This 30-day internship focused on mastering core C++ concepts by building a diverse portfolio of five distinct console applications. The overarching goal was to bridge the gap between basic algorithmic logic and real-world data persistence.

The journey progressed from creating stateless logic programs—like mathematical calculators and dynamic guessing games—into building robust, file-driven systems capable of permanent data storage, such as Hotel and Contact Management software.



Phase 1: Algorithmic Foundations

1. Interactive Calculator

A robust mathematical utility demonstrating fundamental control structures and real-time input parsing.

- ▶ Menu-driven user interface.
- ▶ Handling standard arithmetic operations.
- ▶ Continuous execution loops.
- ▶ Error handling for divide-by-zero scenarios.

2. Number Guessing Game

An application focusing on randomized logic, conditional looping, and interactive user feedback.

- ▶ Implementation of `rand()` logic.
- ▶ Conditional feedback mechanisms ("Too High", "Too Low").
- ▶ Attempt tracking and scoring.
- ▶ Demonstrates core game loop principles.

Phase 2: Data Persistence & Management

3. Hotel Management System

This project marked the transition from stateless memory to persistent fstream file operations, simulating a real-world booking environment.

- ▶ **Room Allocation:** Dynamic assignment and tracking of room availability.
- ▶ **Customer Records:** Collecting and storing customer details in secure text/dat files.
- ▶ **Automated Billing:** Calculating costs based on stay duration and room type.
- ▶ **Data Retrieval:** Ability to search and print invoices from saved history.



Phase 3: Logic & File Integration



4. Tic-Tac-Toe Engine

A two-player console grid game that focuses heavily on multi-dimensional arrays and complex algorithmic win-condition checking (horizontal, vertical, diagonal).

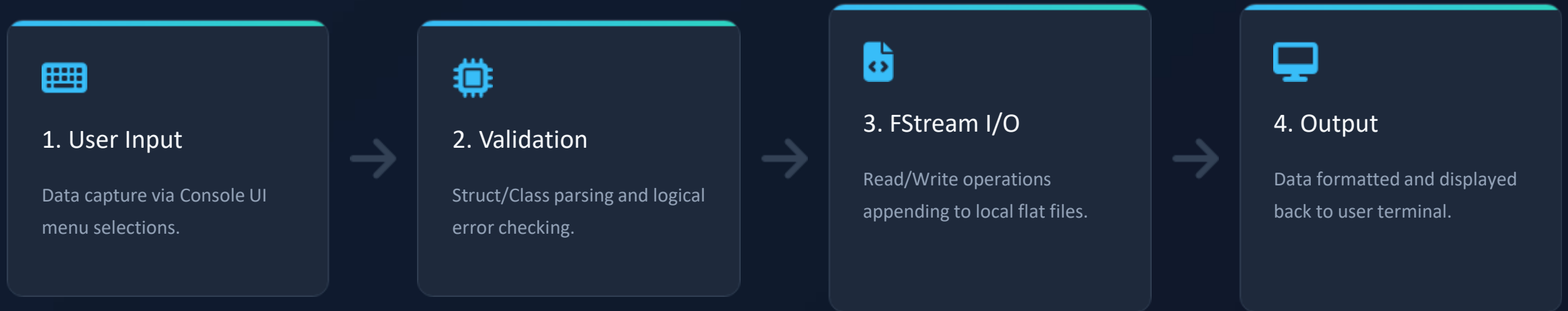


5. Contact Management

A complete CRUD (Create, Read, Update, Delete) application relying on advanced file manipulation. Allows permanent saving, searching, and modification of personal contacts.

Core System Architecture

Standard I/O Flow utilized for the Contact and Hotel Management Systems.



Relative Code Complexity & Scale



The complexity of the projects scaled up significantly, culminating in the Hotel Management system which combined menu UI, algorithmic cost calculation, and persistent file storage.

Thank You

Full project code is available in the GitHub Repository.



github.com/daksh25scs1003004699-blip